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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 3	
TITLE: Method Overloading and Static Members	

Problem Statement

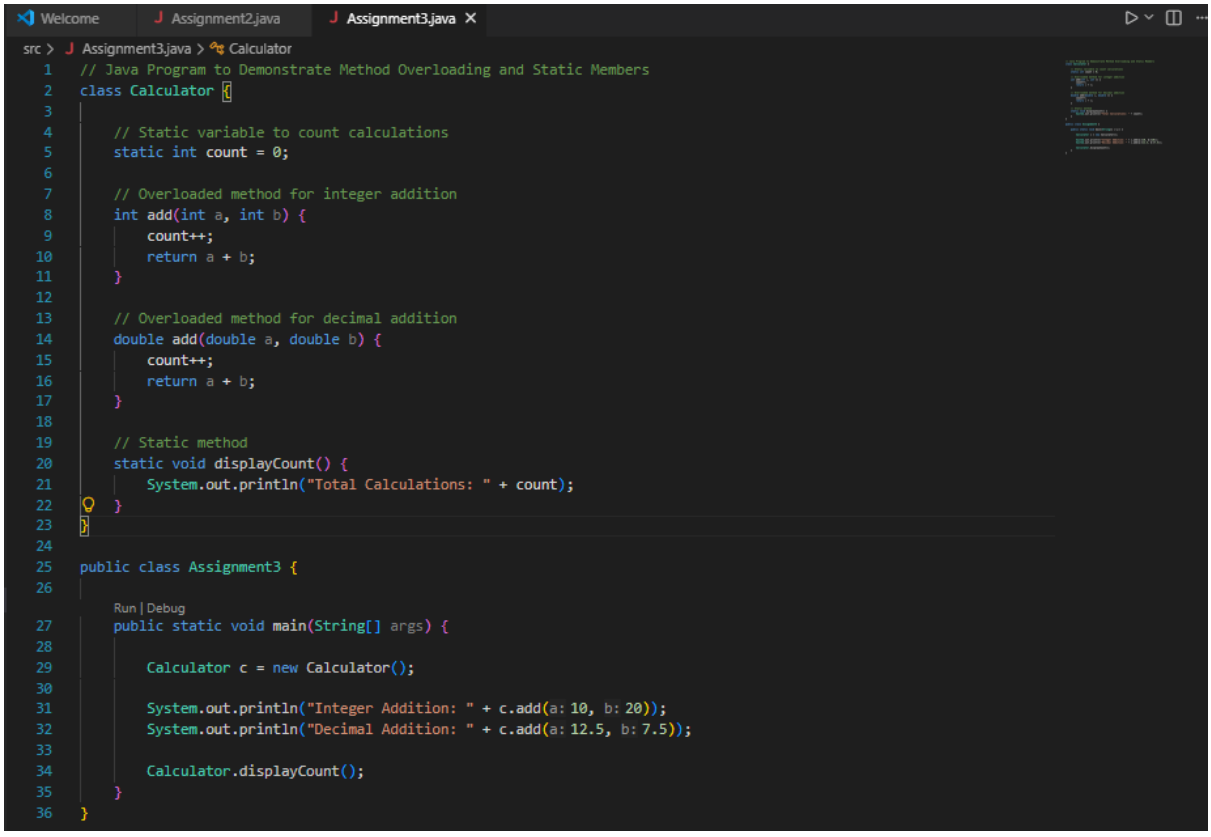
Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Programme



Output

```
PROBLEMS 1  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    Debug: Assignment3

PS C:\Users\RIKISHA\PROGRAMMING\Java>  & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,
address=localhost:63130' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment3
'
Integer Addition: 30
Decimal Addition: 20.0
Total Calculations: 2
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Programme

Welcome | Assignment2.java | Assignment3.java ●

src > Assignment3.java > Restaurant

```java
// Java Program to Demonstrate Method Overloading and Static Members
class Restaurant {

    // Static var to count total orders
    static int totalOrders = 0;

    double bill(double amount) {
        totalOrders++;
        return amount;
    }

    // Takeaway bill
    double bill(double amount, double packingCharge) {
        totalOrders++;
        return amount + packingCharge;
    }

    // Delivery bill
    double bill(double amount, double packingCharge, double deliveryCharge) {
        totalOrders++;
        return amount + packingCharge + deliveryCharge;
    }
    static void displayOrders() {
        System.out.println("Total Orders: " + totalOrders);
    }
}

public class Assignment3 {
    Run | Debug
    public static void main(String[] args) {
        Restaurant r = new Restaurant();

        System.out.println("Dine-in Bill: ₹" + r.bill(amount: 500));
        System.out.println("Takeaway Bill: ₹" + r.bill(amount: 500, packingCharge: 20));
        System.out.println("Delivery Bill: ₹" + r.bill(amount: 500, packingCharge: 20, deliveryCharge: 50));

        Restaurant.displayOrders();
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:
jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:63078' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\User
s\RIKISHA\PROGRAMMING\Java\bin' 'Assignment3'
Dine-in Bill: ?500.0
Takeaway Bill: ?520.0
Delivery Bill: ?570.0
Total Orders: 3
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

The program successfully demonstrated method overloading by defining multiple methods with the same name but different parameters. It also showed the use of

static variables and static methods, which belong to the class and can be accessed without creating an object.